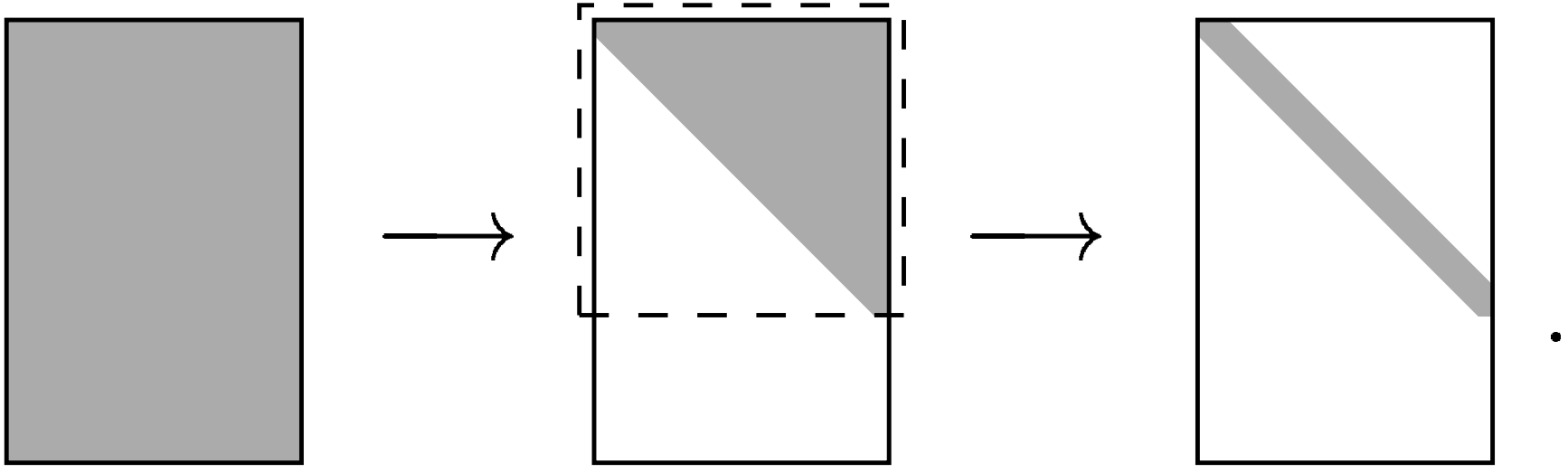
Lawson–Hanson–Chan bidiagonalization



A $\qquad$ Q^*A $\qquad$ U^*Q^*AV